

Michelle Yi Xuan Guo

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Professional Background

Carnegie Mellon Information Systems student pursuing an accelerated M.S. in Information Systems Management and B.S. in Information Systems, with experience in machine learning, causal inference, data engineering, and AI product development. **Canadian citizen** currently studying in the U.S. on F-1 status; open to relocation.

Education

Carnegie Mellon University, B.S. in Information Systems, Aug 2023 – May 2027
Minor in Artificial Intelligence and Architecture
CMU Accelerated Masters Program, MISM - Business Intelligence and Data Analytics Aug 2026 – May 2028

- **Relevant Coursework:** Mobile App Development, Application Design and Development, Database Design and Development, Software Engineering, Computer Systems, Artificial Intelligence, Functional Programming, Statistical Inference, Machine Learning, Matrices and Linear Transformation, Digital Transformation

Experience

Data Scientist, Product Development Team , Summer Atlantic Capital LLC — Remote, New York May 2026 – Present
Project focus: SAC Nexus SustainAI ESG & Impact Intelligence Agent

- Engineered core data infrastructure for SustainAI, developing ingestion pipelines, PostgreSQL schemas, and rule-based/fuzzy entity resolution to standardize ESG disclosures, regulatory filings, and other unstructured data with confidence scoring and source provenance.
- Developed deterministic ESG, impact, and regulatory analytics, including emissions/portfolio aggregation, SDG alignment, and classification logic, with human-in-the-loop review and auditable override APIs.

Causal Inference Research on PSPS impact, CMU — Pittsburgh, PA Summer 2025 – Spring 2026

- Built a Python causal-inference pipeline (pandas, statsmodels) using inverse probability weighting and marginal structural models to estimate PSPS effects on wildfire ignition risk, integrating CalFire ignitions, CPUC outages, and NOAA HRRR weather data.
- Estimated **64.4% lower ignition risk during PSPS events** and identified substantial treatment-effect heterogeneity, motivating counterfactual risk maps to reveal conditions where PSPS effectiveness diverged from expectations.
- Earned a *Travel Award* at the 20th INFORMS DMDA Workshop, **1st Place** in the OR/MS Poster Competition, and runner-up for the *Environmental Undergraduate Research Award*.

Undergraduate Research Fellowship, CMU — Pittsburgh, PA Mar 2025 – Nov 2025

- Developed wildfire-risk models using **GLMs, PyTorch MLPs, and Hawkes processes**, engineering lagged environmental and human-activity features to generate grid-level predictive risk surfaces for California.
- Built a Python data pipeline (xarray, pandas) processing **5 years of hourly weather data across 824 3-km grid cells and 444K+ observations**, integrating wind, temperature, humidity, fuel, and geospatial data for modeling.

Projects

Machine Learning Projects, CMU — Pittsburgh, PA Fall 2025

- Trained neural network models (MLP, CNN, and RNN-style sequence models) in PyTorch, tuning optimization hyperparameters and evaluating generalization performance. Implemented core supervised learning algorithms in Python, including decision trees and linear regression with SGD; achieved > 90% accuracy on synthetic benchmarks and studied overfitting vs. model complexity.
- Implemented value iteration and reinforcement learning agents (Q-learning, policy gradient) for Gridworld, Pacman, and Atari Pong, analyzing learned behaviors and systematic failure modes.

Autonomous Vehicle Technology Scouting Brief — BMW Group, Self-directed research, 2026

- Authored a 15-slide industry scouting brief surveying 15+ AV operators, AI stacks, and Tier-1 suppliers; grounded in 30+ primary sources including peer-reviewed VLM studies, peer-reviewed safety studies (Kusano et al. 2025, *Traffic Injury Prevention*), NHTSA filings, and the UNECE GTR on ADS (Jan 2026).
- Synthesized that AV competition has shifted from sensor hardware to AI software stacks, recommending BMW evaluate end-to-end AI partners, simulation platforms, and complementary China-market stacks beyond the existing Momenta partnership.

Skills

Languages: Python, C, Standard ML (SML), SQL

Tools & Platforms: Git/GitHub, Linux, VS Code, AI-Integrated IDE & Development Tools, PyTorch, pandas, NumPy, statsmodels, scikit-learn, xarray, PostgreSQL, MongoDB, Excel, Microsoft Office Suite

Honors & Awards

- Regional Education Partner scholarship, equal to 30% of full-time tuition
- The 20th INFORMS Workshop on Data Mining and Decision Analytics (DMDA) Travel Award — Fall 2025
- OR/MS Tomorrow 2025 Mini-poster Competition First Place — Fall 2025
- Dean’s List — Fall 2023, Spring 2024, Spring 2026